

## 3.2 Kinematics in 1 Dimension

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs k1–k8

**What this topic covers.** The precise vocabulary of motion in a straight line, motion graphs and what their gradients and areas mean, the constant acceleration formulae, and the use of calculus when acceleration is not constant.

### 1. The vocabulary — and why it matters

MEI distinguishes several pairs that are casually treated as synonyms (spec refs k1–k3).

| Scalar                           | Vector              | Distinction                                                 |
|----------------------------------|---------------------|-------------------------------------------------------------|
| <b>Distance travelled</b>        | <b>Displacement</b> | Total path length vs net change of position, with direction |
| <b>Speed</b>                     | <b>Velocity</b>     | Magnitude only vs magnitude and direction                   |
| <b>Magnitude of acceleration</b> | <b>Acceleration</b> | Never negative vs signed                                    |

**Position** is measured from a fixed origin; **displacement** is the change in position from a starting point.

**Worked example.** A body travels 100 m north then 100 m south in 40 s.

$$\text{average speed} = \frac{200}{40} = 5 \text{ m s}^{-1}, \quad \text{average velocity} = \frac{0}{40} = 0$$

$$\text{average speed} = \frac{\text{distance travelled}}{\text{elapsed time}}, \quad \text{average velocity} = \frac{\text{overall displacement}}{\text{elapsed time}}$$

## 2. Motion graphs

| Graph             | Gradient gives | Area underneath gives |
|-------------------|----------------|-----------------------|
| Displacement-time | Velocity       | No meaning            |
| Velocity-time     | Acceleration   | Displacement          |
| Acceleration-time | —              | Change in velocity    |

- A straight line on a displacement-time graph means constant velocity.
- A horizontal line on a velocity-time graph means constant velocity; a straight sloping line means constant acceleration.
- Area **below** the time axis on a velocity-time graph is displacement in the negative direction.

**Worked example.** A trapezium on a velocity-time graph has parallel sides 4 and 10 and width 6. The displacement is

$$\frac{1}{2}(4 + 10) \times 6 = 42 \text{ m}$$

**Worked example.** A body travels at  $12 \text{ m s}^{-1}$  for 5 s, then decelerates uniformly to rest over the next 4 s. Find the total distance.

Rectangle:  $12 \times 5 = 60$ . Triangle:  $\frac{1}{2} \times 12 \times 4 = 24$ . Total 84 m.

For **distance travelled** rather than displacement, treat any area below the axis as positive. Adding signed areas gives displacement; adding magnitudes gives distance.

## 3. The constant acceleration formulae

With  $u$  initial velocity,  $v$  final velocity,  $a$  acceleration,  $s$  displacement and  $t$  time:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = vt - \frac{1}{2}at^2$$

$$s = \frac{1}{2}(u + v)t$$

$$v^2 = u^2 + 2as$$

Each omits one variable, so choose the one that avoids the quantity you neither know nor want. The fifth omits  $t$ ; the first omits  $s$ .

These apply **only** when the acceleration is constant. If acceleration is given as a function of time, they are invalid and calculus is required.

**Worked example.** A particle starts from rest and accelerates at  $2 \text{ m s}^{-2}$  for 5 s.

$$v = 0 + 2(5) = 10 \text{ m s}^{-1}, \quad s = 0 + \frac{1}{2}(2)(25) = 25 \text{ m}$$

**Worked example.** A car travelling at  $20 \text{ m s}^{-1}$  decelerates uniformly at  $4 \text{ m s}^{-2}$  to rest.

$$0 = 20 - 4t \implies t = 5 \text{ s}, \quad s = \frac{1}{2}(20 + 0)(5) = 50 \text{ m}$$

**Worked example.** A body has  $u = 5 \text{ m s}^{-1}$  and accelerates at  $3 \text{ m s}^{-2}$  over 10 m.

$$v^2 = 25 + 2(3)(10) = 85 \implies v = 9.22 \text{ m s}^{-1} \text{ (3 s.f.)}$$

## Vertical motion under gravity

Taking  $g = 9.8$  and being consistent about the positive direction:

**Worked example.** A stone is dropped from rest. After 2 s,

$$v = 9.8(2) = 19.6 \text{ m s}^{-1}, \quad s = \frac{1}{2}(9.8)(4) = 19.6 \text{ m}$$

**Worked example.** A stone falls from rest through 45 m. Find the time.

$$45 = \frac{1}{2}(9.8)t^2 \implies t^2 = 9.184 \implies t = 3.03 \text{ s (3 s.f.)}$$

**Worked example.** A ball is thrown vertically upwards at  $14.7 \text{ m s}^{-1}$ . Find the maximum height.  
At the highest point  $v = 0$ :

$$0 = 14.7^2 - 2(9.8)s \implies s = \frac{216.09}{19.6} = 11.025 \text{ m}$$

Fix the positive direction at the start and keep it. If upwards is positive, then  $a = -9.8$  throughout — including on the way up, at the top, and on the way down. The acceleration never changes; only the velocity does.

## 4. Calculus for variable acceleration

When acceleration is not constant, differentiate to go down the chain and integrate to go up (k5):

$$v = \frac{ds}{dt}, \quad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$v = \int a \, dt, \quad s = \int v \, dt$$

**Worked example.** For  $s = t^3 - 3t$ , find the velocity at  $t = 2$ .

$$v = 3t^2 - 3 \implies v(2) = 12 - 3 = 9 \text{ m s}^{-1}$$

**Worked example.** A particle has  $a = 6t$  and velocity  $2 \text{ m s}^{-1}$  at  $t = 0$ .

$$v = \int 6t \, dt = 3t^2 + c, \quad v(0) = 2 \implies c = 2$$

$$v = 3t^2 + 2$$

The constant of integration is fixed by the initial condition. Omitting it, or forgetting to apply the condition, is the standard error in these questions.

## Distance travelled when the direction reverses

**Worked example.** For  $s = t^3 - 6t^2 + 9t$ , find the times at rest and the total distance travelled from  $t = 0$  to  $t = 4$ .

$$v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \implies t = 1 \text{ and } t = 3$$

Evaluating the displacement at the key times:

$$s(0) = 0, \quad s(1) = 4, \quad s(3) = 0, \quad s(4) = 4$$

The particle moves forward 4 m, back 4 m, then forward 4 m again. So the **distance travelled** is 12 m, while the **displacement** is only 4 m.

Whenever the velocity changes sign inside the interval, split at those instants and add the magnitudes. Integrating straight through gives the displacement, not the distance — and the question usually wants the distance.

## 5. Relative velocity in one dimension

The velocity of  $A$  relative to  $B$  is  $v_A - v_B$ . Two cars travelling the same way at  $30$  and  $20 \text{ m s}^{-1}$  have a relative velocity of  $10 \text{ m s}^{-1}$  — which is why the gap closes slowly despite both moving quickly.

## 6. Common pitfalls

- Using the suvat formulae when the acceleration is not constant.
- Changing the sign of  $g$  partway through a vertical motion problem.
- Giving displacement when the question asks for distance travelled.
- Omitting the constant of integration, or not applying the initial condition.

- Confusing the gradient and the area of a velocity-time graph.
- Assuming acceleration is zero at the highest point of a vertical throw — the velocity is zero, not the acceleration.
- Reporting a negative distance.